



zip in Python

- The `zip` function takes two or more iterables (lists, tuples, etc.) and **groups them element-wise into tuples**.
- The result is an iterable of tuples where the i-th tuple contains the i-th elements from each input iterable.
- The length of the result is equal to the shortest input iterable.

Example:

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```
names = ["Alice", "Bob", "Charlie"]
ages = [25, 30, 35]

zipped = zip(names, ages)

print(list(zipped))

# Output: [('Alice', 25), ('Bob', 30), ('Charlie', 35)]
```



map in Python

- The `map` function applies a function to each element of one or more iterables and returns an iterable with the results.
- It's used to transform data in a functional style.

Basic example:

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```
numbers = [1, 2, 3, 4]
```

```
squared = map(lambda x: x**2, numbers)
```

```
print(list(squared))
```

```
# Output: [1, 4, 9, 16]
```



Combining zip and map

Imagine you have two lists and want to sum their corresponding elements:

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```
a = [1, 2, 3]
b = [4, 5, 6]

result = map(lambda x: x[0] + x[1], zip(a, b))

print(list(result))
# Output: [5, 7, 9]
```

You can also use unpacking for clarity:

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```
result = map(lambda x, y: x + y, a, b)

print(list(result))
# Output: [5, 7, 9]
```